Why is Denmark so eager to maintain its Peg? Analysis of the Danish Exchange Rate Regime

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1. Introduction

Despite being a member of the European Union, Denmark refused to adopt the Euro as their own currency in a referendum in 2000. Since 53 percent of the Danish Population voted against the transition to the Euro, the Krone was retained and pegged to the Euro for more than 20 years, up until today (Abildgren, 2010). Even though the peg was challenged several times in the past, the governor of the Danmarks Nationalbank Lars Rohde is determined to "do whatever it takes to maintain the peg" (Milne, 2015). The commitment to the peg is reflecting the conviction of its favourable consequences for the Danish economy. This paper aims to ascertain those benefits in order to answer the guiding question of this paper: "Why is Denmark so eager to maintain its peg?"

To sufficiently answer this research question, the paper will explore how the decision of a fixed exchange rate regime impacted a thorough selection of economic variables. The paper will provide an argumentation based on empirical data and macroeconomic models. Firstly, the paper will outline a historical overview of Denmark's exchange rate policy. Secondly, it will examine the aims of the regime and benefits associated with it, thereby introducing Denmark's inflation rate, trade patterns and external wealth position. Thirdly, a critical analysis of the fixed exchange rate regime will be provided, by examining the economic developments around the financial crisis. As a conclusion, the paper will evaluate the findings to answer the research question and further to weigh the arguments in favour and against the peg.

2. Historical Background

Denmark has been fixing its currency, the Danish Krone (DKK), for most of its time. When established at the end of the 19th century, the DKK was part of the gold standard. It joined the Bretton-Woods system after World War II and was thus pegged to the US-dollar until the system collapsed in the 1970s. Denmark entered the first European Exchange Rate Mechanism (ERM) and pegged its currency to the European Currency Unit (ECU) in 1982. Since 1999 with the establishment of the Euro, Denmark fixed its currency to the Euro within the system of the second European Exchange Rate Mechanism (ERM II). Thus, Denmark has been able to keep up a

successful fixed exchange rate for more than three decades of which the last 20 years only showed minor fluctuations, illustrating the commitment of the Danmarks Nationalbank (DNB) to the peg (Abildgren, 2010).

2.1. European Exchange Rate Mechanism 1979-1999

After the Bretton-Woods system collapsed, the European Economic Community introduced the ERM in 1979 to regain monetary stability in Europe by stabilising exchange rates within Europe. The ERM only partly succeeded in its first years, and Denmark devalued the DKK multiple times in those years. It was not until 1982 that the Danish government decided to take the peg more seriously due to high inflation rate. The DKK was pegged to the ECU, a basket of European Currencies. However, since other major European currency directly pegged to the Deutsche Mark, the ECU represented the German currency at that time. In the 1990s, many European countries had to leave the ERM during the exchange rate crises, while Denmark continued to commit to the ERM under a wider fluctuation band (Abildgren, 2010).

2.2. European Exchange Rate Mechanism II 1999-Today

The era of the ERM ended in 1999 with the establishment of the Euro. Since then, the DKK has been pegged to Euro within the ERM II, which allows the DKK to fluctuate by +/- 2.25 percent to the Euro. The actual exchange rate, however, never fluctuated by more than 1 percent in the last 20 years. Normally, all countries that have been in the ERM II joined the Euro eventually. Denmark, however, negotiated an opt-out clause of not having to join the Euro and is in addition the only permanent country in the mechanism since 1999 (Abildgren, 2010).

3. Benefits of the Peg

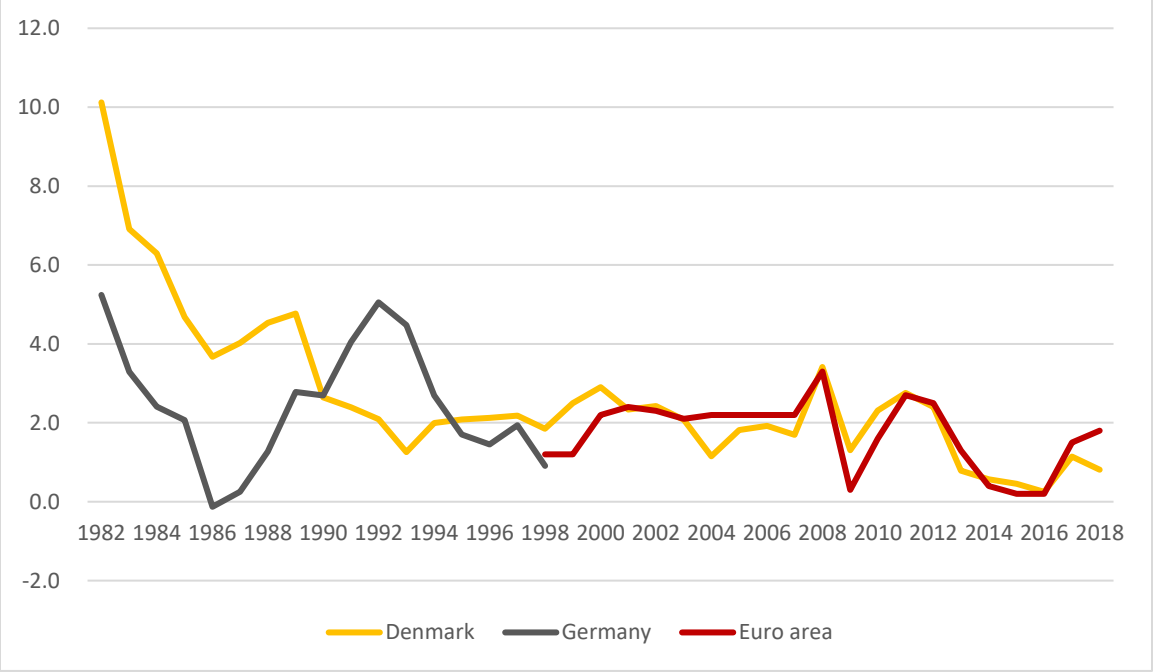
The long history of Denmark's fixed exchange rate and the strong commitment of the DNB to sustain it, suggests significant benefits associated with the application of this regime. As any of the other two nominal anchors, a fixed exchange rate supports price stability. In addition to that, a fixed exchange rate contributes to an increase in international trading, especially between the

central country and the non-base country. In the case at hand, between the Euro Area and Denmark. As a result, the peg permits to involve a country more deeply in the international market and the resulting increase in credibility amongst the trading partners enhances the frequency of trade.

3.1. Price Stability

One of the reasons why the DNB is so determined to keep a fixed exchange rate is to maintain low and stable inflation. Figure 1 represents inflation as a percentage of GDP for Denmark, Germany and the Euro Area.

Figure 1. Inflation Rate (Consumer Prices)



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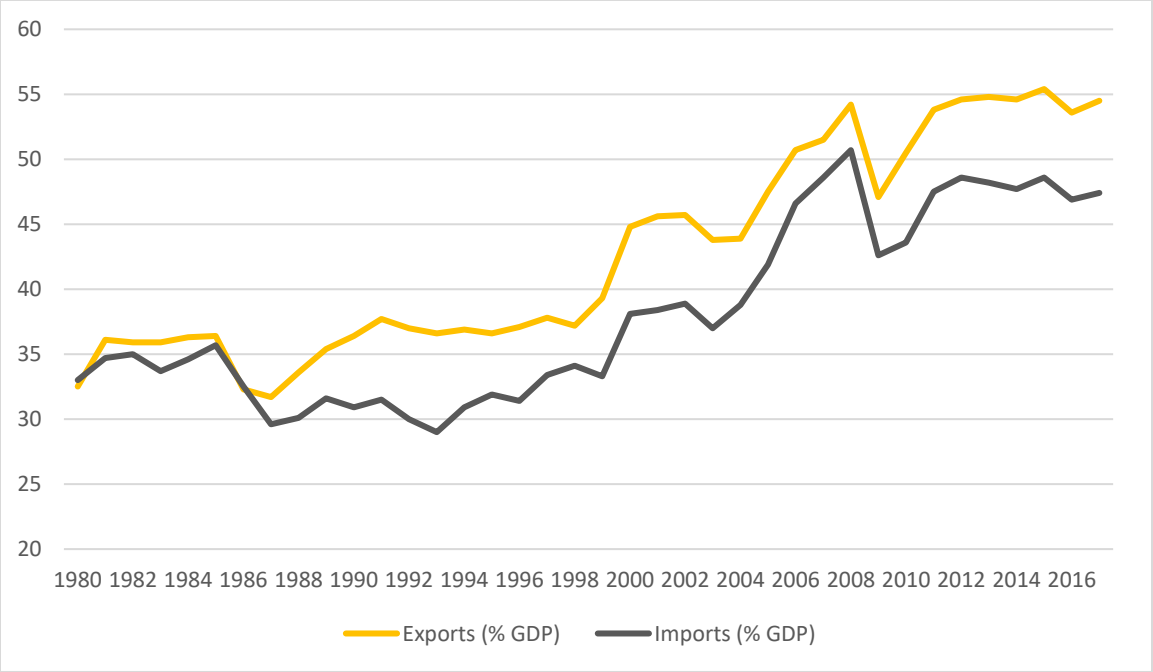
Denmark was initially pegged to the German Mark, mirrored by the similitude of the patterns followed by both countries' inflation rate. Nevertheless, Denmark had a very high inflation rate, when compared to Germany. However, since 1999, when Denmark established the peg to the euro, its inflation rate has been more stable and lower ever since. Furthermore, throughout the last decade Denmark's inflation rate converged and now fluctuates with minor deviation from the ECBs inflation target of 2 percent. The similitude between the price level of Denmark and the Euro Area

has provided Denmark with a high degree of competition, especially in the European Area. As a consequence of increased competitiveness and low transaction costs, the frequency of trade drastically increased between the two economies. Therefore, the fixed exchange rate regime, upheld by the DNB, has achieved its objective of preserving inflation stability and contributed to an increase in trade within the Euro Area through price stability.

3.2. Trade Position

Regarding Denmark's role in the international market, it is the 38th largest exporter in the world (OEC, 2017). Its top five export destinations are Germany, Sweden, United States, United Kingdom and Norway. It is deducible that the vast majority of its international trade is occurring within Europe as 70 percent of Denmark's exports flow into European countries and further 80 percent of its goods are imported from European countries (OEC, 2017). The geographic proximity facilitated extensive trade within the Eurozone and was further amplified by stable prices due to the exchange rate regime.

Figure 2. Denmark's Export and Import (% of GDP)



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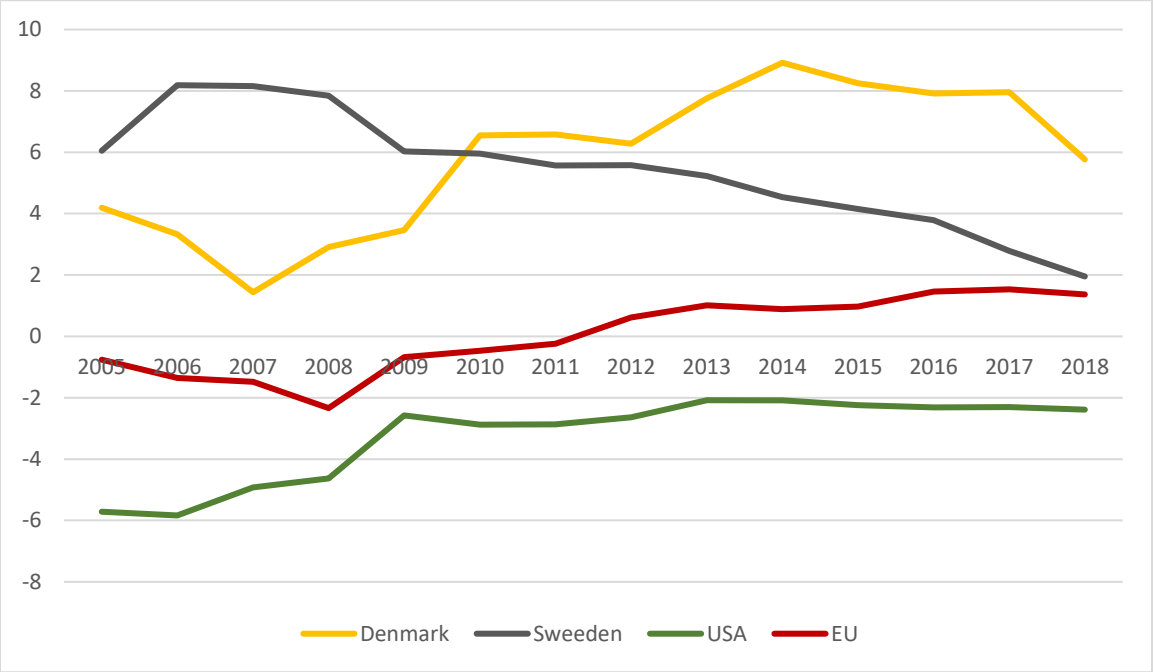
In addition, Denmark has maintained a significant trade surplus for a long time since 1987. This fact is visualized in Figure 2, which illustrates the development of Denmark's total imports and exports as a percentage of GDP. As portrayed by the graph, total exports always exceed total imports. Once again, since 1999 both exports and imports have increased more steeply, in the sense that the peg has positively contributed to the trade balance in Denmark's economy.

The Danish total trade to GDP ratio is above 100 percent and the current export to GDP ratio is 53 percent, thereby demonstrating Denmark's high dependency on international trade. The fact that exports contribute to more than 50 percent to the GDP highlights the importance of price stability. The tendency of the DKK to appreciate against the Euro threatens the Danish Economy, which extensively relies on exports as a source of GDP (Danmarks Nationalbank, 2019). If the DKK strengthens against the Euro, European countries would find it more difficult to import from Denmark due to the increase in prices and henceforth diminish the demand for Danish exports. The corresponding absence of competitiveness is resulting in a decrease of the Danish trade balance, which is mirrored in a decrease in its GDP and furthermore in an increase of unemployment. Consequently, the fixed exchange rate has not only contributed to the extension of the Danish trade balance but also avoids possible adverse effects on employment and GDP due to price volatility.

3.3. External Wealth

A further economic variable explored by the paper is net external wealth- more specifically its accumulation, denomination and the connection to the exchange rate. Denmark's current account surplus helps to explain its position in the foreign market as well as its accumulation of high external wealth. The current account is an indicator of the amount of transactions between a country and the rest of the world. The subsequent graph, Figure 3, compares the current account of Denmark, the European Area and the United States.

Figure 3. Current Account (% of GDP)



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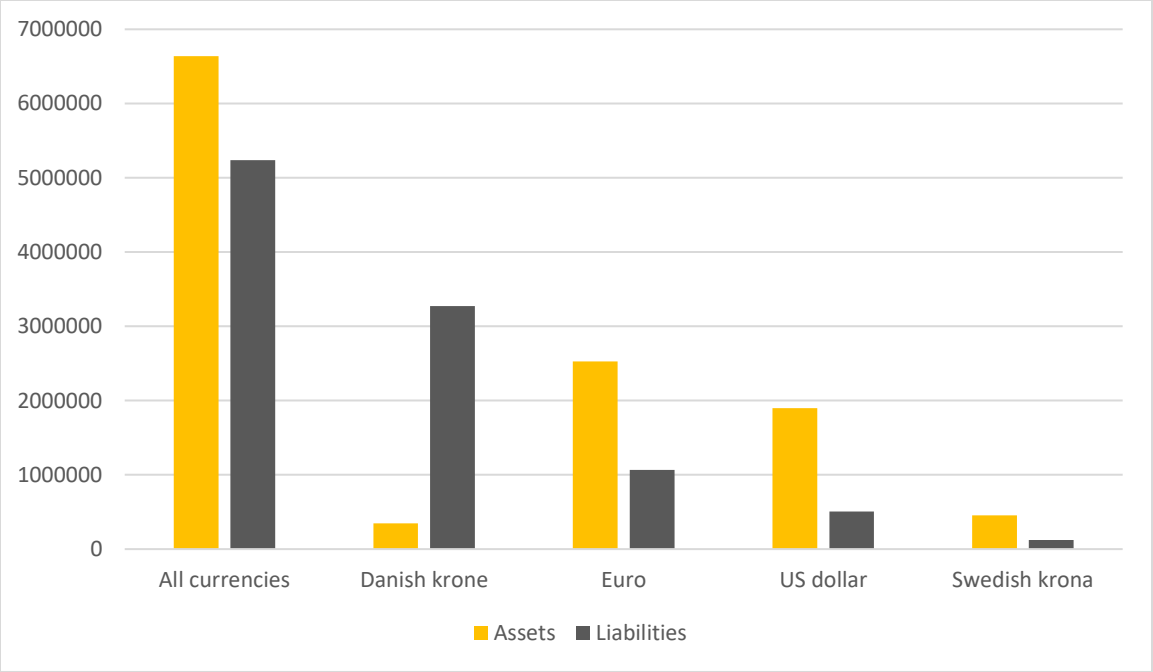
Denmark's current account has been in surplus for more than a decade and is, with difference, above the Euro Area and the United States. This validates the extent and high involvement in foreign trade, which is positively contributing to the health of Denmark's economy. The current account can be further defined as the difference between savings and investment: CA = S – I (Feenstra and Taylor, 2017). Denmark's positive current account indicates that domestic savings are higher than domestic investment in the country. The excess of domestic savings results in a high level of capital outflows, thereby enabling Denmark as a creditor country and enhancing its external wealth position.

Denmark succeeded to gain a strong international investment position, with net external wealth corresponding to 65 percent of its GDP (Danmarks Nationalbank, 2018). Over the past the DKK revealed the tendency to strengthen against the Euro, which constitutes to the conclusion of the Danish currency being undervalued under the peg (Danmarks Nationalbank, 2019). In order to maintain it's hard peg, Denmark needed to counteract such appreciations by increasing its foreign euro reserves to put an upward pressure on the Euro. The decision of letting the DKK float against

the Euro would most likely lead to an unprecedented jump in the value of the DKK. This finding has further important implications for the determination of the net external wealth position of Denmark.

Figure 4. Denmark's foreign assets and liabilities (in Million DKK)



Note. From: *Denmark's International Investment Position* (database). Copyright 2019 by Danmarks Nationalbank

Figure 4 depicts the Division of Denmark's foreign assets and liabilities and their denomination in different currencies. The first two bars demonstrate that foreign assets are exceeding foreign liabilities and thus verify Denmark's international investment position as a net creditor. Furthermore, the graph highlights the fact that a large portion of Denmark's liabilities are denominated in the DKK. Henceforth, representing Denmark's ability to issue domestic and external debt in their own currency. Due to its long-term commitment of price stability Denmark has proven to be a credible borrowing country and its government debt enjoys the highest credit risk grade of AAA (Danmarks Nationalbank, 2019). Equally important as the denomination of Denmark's liabilities in DKK, is the denomination of Denmark's assets in Euros. The fact that most of Denmark's assets are denominated in Euros can be traced back to its obligation to invest in the Euro if its own currency strengthens.

This paper analyses external wealth with the help of the following equation:
$W = (A_{DKK} + EA_{€}) - (L_{DKK} + EL_{€})$ (Feenstra and Taylor, 2017). In the case at hand external wealth is largely composed of Denmark's external assets $A_{€}$ denominated in Euros and external liabilities L_{DKK} denominated in DKK. We find external assets $A_{€}$ in the first term of the equation, representing the majority of assets. The external assets can be converted to DKK with the help of the nominal exchange rate E. External liabilities L_{DKK} enter the equation in the second term without the necessity of being converted, thus independent of the nominal exchange rate E.

In the scenario in which the DNB decides to let the DKK float, it will appreciate against the Euro, reflecting a decrease in the exchange rate. As a result, the first term of the equation, experiences a large decrease in value as the decreasing exchange rate affects the majority of the external assets. By contrast, most of the foreign liabilities are denominated in DKK and are therefore mostly unaffected by the decrease of the Exchange rate. Succinctly, the value of the assets drops by a substantial amount, which is only partly offset by a slight decrease in the value of foreign liabilities. The decrease in the Exchange rate leads to a decrease in the net external wealth of Denmark. This preference of keeping the exchange rate fixed instead of letting it float is often referred to as the "The fear of floating" and demonstrates the case in which a floating exchange rate is likely to decrease the net external wealth position (Feenstra and Taylor, 2017). The analysis of the determination of the net external wealth position provided another argument in favour of the currency peg.

4. Sacrifices of the Peg

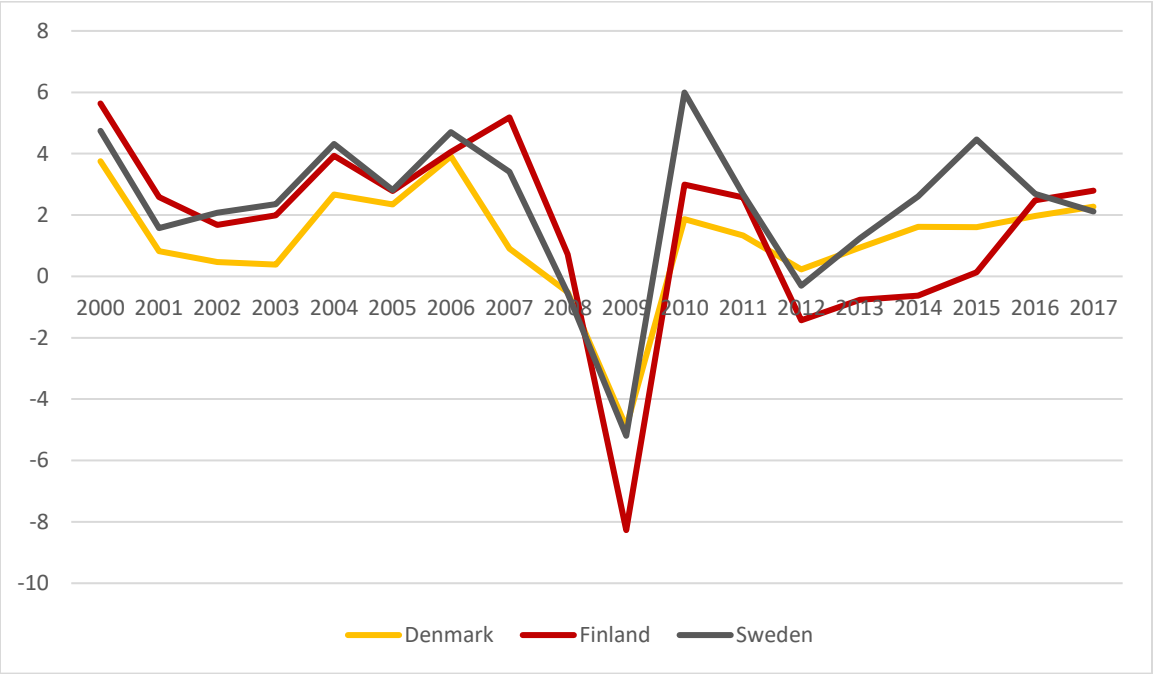
Despite bringing price stability and trade openness, the fixed exchange rate comes with one major drawback. Highlighted by the so-called Trilemma Triangle, a country that fixes its exchange rate and does not impose any capital controls must forfeit its monetary autonomy (Feenstra and Taylor, 2017). Thus, the DNB is limited to mirroring the actions of the European Central Bank (ECB) to ensure that the exchange rate does not fluctuate and that free capital flows are provided.

To explore the consequences of this implication, the analysis focuses on two scenarios, in which the exchange rate regime has been criticized. Firstly, we investigate the years following the financial crisis in 2009, where the fixed exchange rate was blamed for Denmark's slow recovery compared to its neighbouring country Sweden, which has a floating exchange rate. Secondly, Anderson (2014) argued that Denmark was not able to control its overheating economy prior the crisis due to the inability of enforcing independent monetary policies. Both arguments are analysed with the help of data and economic models.

4.1. Financial Crisis Recovery

Denmark experienced a slow recovery from the crisis in comparison to Sweden, whose currency is floating to the Euro. This can be seen in Figure 5, which depicts the real GDP growth rate of Denmark, Sweden and Finland from 2000 onwards.

Figure 5. Real GDP Growth Rate (%)

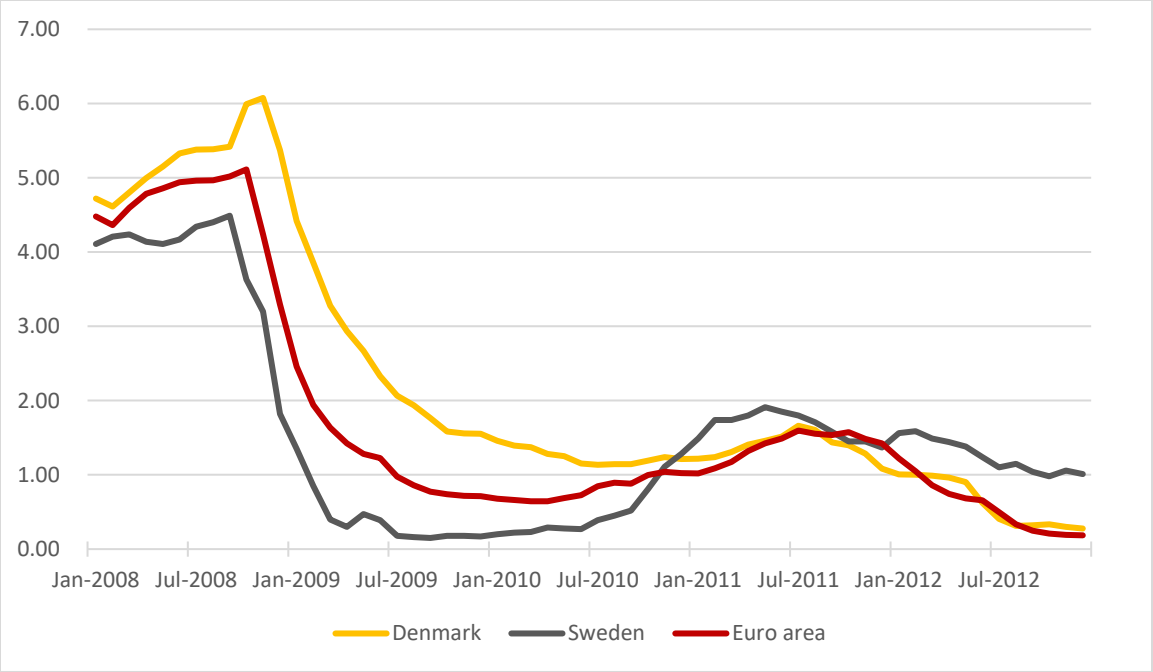


Note. From: *World Economic Outlook* (database). Copyright 2018 by International Monetary Fund

Multiple observations that are important for our analysis can be concluded from the graph. The graph visualizes that Denmark's GDP growth rate indeed falters especially compared to Sweden's growth rate. However, the financial crisis in 2009, however seems to have affected the different

markets similarly and accordingly represents a symmetric shock in Europe. Moreover, Sweden's GDP growth rate has also been greater before the crisis, which indicates that both economies differ by other variables than the exchange rate, despite being akin Nordic countries.

Figure 6. Short-Term Interest Rate



Note. From: "Finance", *Main Economic Indicators* (database). Copyright 2019 by OECD

Still, Denmark's exchange rate policy is criticized for the slow recovery, since it imposes limitations on the DNB to enforce monetary easing. Monetary easing is a policy in which the interest rate is lowered to stimulate the economy and increase its output. Figure 6 displays the short-term interest rates of Denmark, Sweden and the Euro Area. As expected, Sweden followed a more aggressive monetary easing policy than Denmark. The DNB reacts to interest rate changes the ECB imposes, resulting in a slow monetary easing policy that might be insufficient stimulus for the Danish economy. Regardless of the crisis being a seemingly symmetric shock, European economies were still affected slightly different. As illustrated by Figure 5, Finland was hit subsequently harder by the crisis than the other Nordic countries.

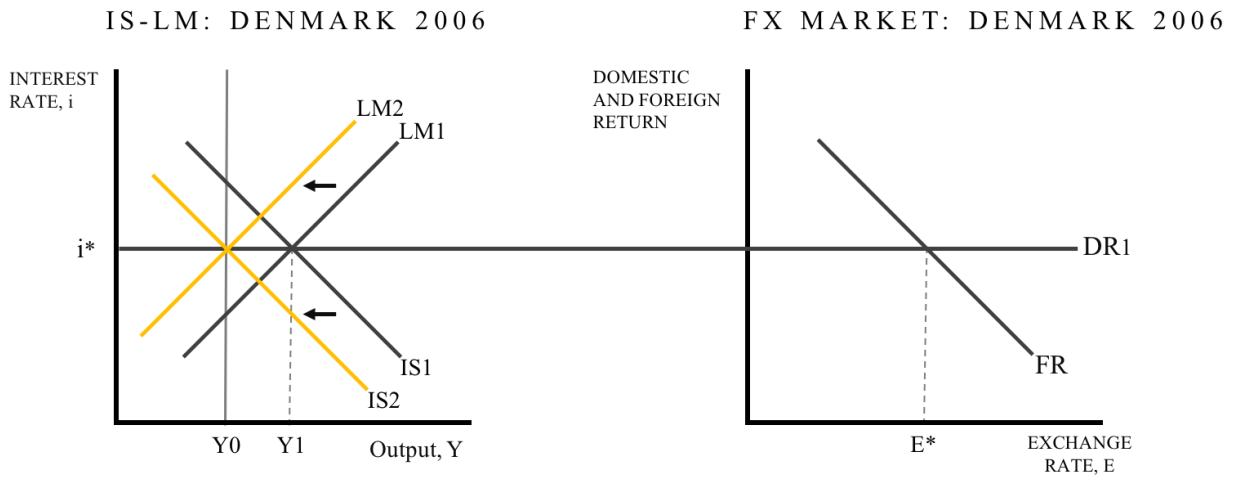
Therefore, whichever interest rate the ECB chose, it could not reflect the optimal policy decision for all countries in the Euro Area, and by that most likely did not reflect Denmark's optimal approach. In that way, Denmark's exchange rate policy possibly restricted the DNB of following

its optimal monetary response to the crises, which could partially explain the slower recovery. However, there are many more variables that are not considered in this analysis, and as stated before Sweden's generally higher growth rate than Denmark implies further differences between both economies.

4.2. Overheating Economy

A country specific development prior to the crisis was Denmark's strong overheating economy, in a manner that the actual output exceeded the potential output of the economy. Denmark's output gap, meaning the difference of GDP and potential GDP, was highest in the years preceding the crisis with actual GDP being 3% higher than the potential GDP for at least the last 30 years (International Monetary Fund, 2018). This overheating was especially visible in the rise of housing prices in Denmark. Anderson (2014) shows the correlation that countries with a bigger housing bubble suffered a heavier recession. Given that Denmark had one of the biggest housing bubbles in Europe at that time (Anderson, 2014), the overheating economy presents another explanation for the slow recovery.

Figure 7. IS-LM-FX Model of Denmark 2006



Note. This model represents an overheating economy such as Denmark's in 2006 from a theoretical approach and does not provide empirical data.

Figure 7 represents an IS-LM-FX diagram of Denmark's economy in 2006. This model aims to demonstrate how the absence of monetary autonomy constitutes to an overheating economy. The

initial output is at Y1, thereby exceeding the potential output of Y0, while the exchange rate is fixed at E*. A country with monetary autonomy could antagonize an overheating economy by simply imposing a monetary contradiction to lower its output therewith appreciating its currency. Denmark, on the other hand, is obliged to maintain its peg and is restricted to the use of monetary contradiction with a simultaneous fiscal tightening to reduce its output. This policy mix is represented in Figure 7 by a leftward shift of both the IS and the LM curve. In that manner, output returns to its potential output Y0 while the exchange rate remains fixed at E*. The DNB acknowledged that dampening the economy before the crisis hit would have been more beneficial and was in favour of a fiscal tightening, which was however never enforced (Danmarks Nationalbank, 2006). Succinctly, with monetary independence the DNB would have been enabled to react to the overheating with a monetary contradiction and consequently might have moderated the depth of the recession in the following years.

5. Comparison/Evaluation

To answer the research question why Denmark is so committed to the peg, the evaluation of the findings must be conducted in two steps. By summarising and weighing the findings, the paper provides an answer to the reasoning behind the decision to peg the currency instead of a floating exchange rate regime. Besides letting the DKK float, Denmark also holds the option of adopting the Euro. Denmark demonstrated serious commitment to the peg, therefore further insight is provided to answer why Denmark is eager to maintain its peg instead of joining the Euro.

5.1.Fix vs Floating

Initially, Denmark decided to fix its currency to decrease inflation. This idea of price stability is still relevant today and having the exchange rate as a nominal anchor has been successful at maintaining this goal. Additionally, since Denmark's economy relies heavily on international trade, especially within Europe, the fixed exchange rate to the Euro encourages Denmark's position of trade openness. This resulted further in an accumulation of external wealth over the last decade, in which most of the foreign liabilities are denominated in DKK, while assets are denominated in Euro. Due to the trend of the DKK to appreciate to the Euro, Denmark endures a

reverse fear of floating. Furthermore, the DNB has been proven competent of keeping up a fixed exchange rate regime for more than 30 years, which earned them a reputation of high credibility. Nevertheless, those benefits come with the sacrifice of monetary autonomy. The paper introduced two examples in the context of the financial crisis and analysed whether the absence of monetary autonomy supposedly led to negative economic developments. Not being able to control an overheating economy without fiscal policy and the barriers on monetary easing were in the focus of the analysis. However, due to many lurking variables, the analysis fails to pinpoint the exact contribution of the exchange rate regime to the amplified adverse effects of the crisis. Considering Denmark's current trade and external wealth position, while still accounting for the costs of the monetary autonomy sacrifice, the benefits of the fixed exchange rate outweigh the cost of the regime choice.

5.2.Fix vs Euro

The previous section of the paper concluded from an economic point of view that the favourable consequences of the currency peg outweigh the cost of a no monetary autonomy regime. Denmark, as a member of the ERM II is essentially in preparation for a transition to the Euro. In 2000, 53 percent of the Danish population voted against the adoption of the Euro. However, the government still followed a pro-euro political establishment to transition in the future (Sjolin, 2017). The belief that Denmark's opt out clause and hence it's possibility to deviate from the ECBs policy only provides benefits for the country was challenged several times in the past, as Denmark had to endure severe speculative attacks. The last speculative attack, occurring in 2015, made it more than clear, that the opt out clause provides a ground for speculation and thereby imposes a threat of adverse effects due to Denmark's obligation to counteract the speculative capital inflows (Jolly, 2015).

Over the last decades, Denmark's nominal exchange rate never deviated by more than one percent from the ECBs policies and furthermore the DNB made it very clear, that it does not intend to do so in the future. Consequently, the potential benefit of the fluctuation band instead of the adoption of the Euro is being eroded, whereas the risk of speculative attacks remains. Henceforth, the logical consequence would be the transition to the Euro, which is consistent with the suggestion of the governor of the DNB Lars Rohde. Nevertheless, the ultimate decision is made by the public, which might be driven by emotional rather than economical reasoning. Here, the increasing trend of

national interest decreases the likelihood of a pro-euro decision in a second referendum. However, the analysis demonstrated an increase of Denmark's market integration since 2000, which strengthens the argumentation in favour of the adoption of the euro.

6. Conclusion

This paper aimed to analyse Denmark's decision to peg its currency and to back up the argumentation with empirical data and macroeconomic models. Firstly, the paper provided an argumentation in favour of the peg, taking into account Denmark's inflation rate, trade patterns and its international wealth position. Figure 1 demonstrated the convergence of Denmark's inflation rate to the European Central Banks target of two percent. Furthermore, the effect of price stability on trade patterns was explored and demonstrated how the peg enhanced Denmark's international trade position and underlined the reliance of exports as a source of GDP. In addition to that, the current account of Denmark was hinting towards a strong net external wealth position and was later verified by the analysis of Denmark's denomination of foreign assets and liabilities. Both the heavy reliance on exports as well as Denmark's division of foreign assets and liabilities highlighted the importance of a fixed exchange rate, especially towards the Euro Area, its major trading and investment partner.

The paper then inspected the adverse effects of the fixed exchange rate regime in the context of the financial crisis. The analysis demonstrated that Denmark's slow recovery from the financial crisis and the overheating economy preceding the crisis might be the outcome of Denmark's sacrifice of monetary autonomy. However, due to the limitations of the models and variables used, the paper failed to pinpoint the exact contribution of the exchange rate on the adverse effects. Thus, the paper concludes that the benefits of pegging the currency outweigh the equivocal costs of sacrificing monetary autonomy. Furthermore, the paper suggests the full transition to the Euro, however questions the likelihood of a pro-euro referendum decision.

7. References

Abildgren, K., Andersen, B. N., & Thomsen, J. (2010). *Monetary history of Denmark 1990-2005*. Danmarks Nationalbank. Retrieved from: https://www.nationalbanken.dk/en/publications/Documents/2010/12/Monetary_History_Denmark_web.pdf

Andersen, T.B., & Malchow-Møller, N. (2014). *Denmark's fixed exchange rate regime and the delayed recovery from the Global Financial Crisis: A comparative macroeconomic analysis.* University of Southern Denmark. Retrieved from: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2435859

Danmarks Nationalbank (2018). *Denmark's International Investment Position, 3rd Quarter 2018.* Retrieved from: http://www.nationalbanken.dk/en/statistics/find_statistics/Documents/Denmarks%20international%20investment%20position/Denmarks%20international%20investment%20position%2020181221.pdf

Danmarks Nationalbank (January 2019). *Central Government Debt.* Retrieved from: http://www.nationalbanken.dk/en/governmentdebt/centralgovernment_debt/Pages/default.aspx

Danmarks Nationalbank (January 2019). *The Foreign-Exchange Reserve.* Retrieved from: http://www.nationalbanken.dk/en/monetarypolicy/foreign_exchange_reserve/Pages/default.aspx

Feenstra, R. C. & Taylor, A. M. (2009). *International Economics* (4th internat. ed.). New York. Worth Publishers.

International Monetary Fund (2018), *World Economic Outlook* (database). Retrieved from: https://www.imf.org/external/pubs/ft/weo/2018/02/weodata/index.aspx

Jolly, D. (2015, February 12). Concern in Denmark as Its Currency's Peg to the Euro Is Strained. *The New York Times*. Retrieved from: https://www.nytimes.com/2015/02/13/business/international/concern-in-denmark-as-its-currencys-peg-to-the-euro-is-strained.html

Milne, R. (2015, February 6). Danish central bank fiercely defends currency peg. *Financial Times*. Retrieved from: https://www.ft.com/content/d3c385f6-adc6-11e4-919e-00144feab7de

Observatory of Economic Complexity (2017). *Denmark.* Retrieved from: https://atlas.media.mit.edu/en/profile/country/dnk/

OECD (2019). Balance of payments BPM6, *Main Economic Indicators* (database). Retrieved from: https://doi.org/10.1787/data-00706-en

OECD (2019). Finance, *Main Economic Indicators* (database). Retrieved from: https://doi.org/10.1787/data-00043-en

OECD (2019). National Accounts at a Glance, *OECD National Accounts Statistics* (database). Retrieved from: https://doi.org/10.1787/data-00369-en

OECD (2019). Prices: Consumer prices, *Main Economic Indicators* (database). Retrieved from: https://doi.org/10.1787/0f2e8000-en

Sjolin, S (2017, December 11). The Game of Krones: Inside Denmark's battle to defend its 35-year-old currency peg. *Market Watch.* Retrieved from: https://www.marketwatch.com/story/the-game-of-krones-inside-denmarks-battle-to-defend-its-35-year-old-currency-peg-2017-12-08

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